Electromagnetic Waves Inside a Material

Light (oscillator)
frequency $\omega = ck$

Dipoles in medium (oscillators)
frequency ω_0

Coupling
$E \sim P$ (Maxwell)
$P \sim E$ (Newton)

↓

Frequency of coupled modes

$\frac{\omega^2}{c^2}\varepsilon(\omega) = k^2$ polariton dispersion in bulk medium

Before being coupled, we have two separate oscillators. The electromagnetic field has

$$\omega_{ph}(k) = \frac{ck}{\sqrt{\varepsilon_\infty}}$$

while material has optical polarization resonance around ω_T. The electric field drives the polarization $E \to P$ and the polarization radiates back $P \to E$. Neither can oscillate independently once they interact strongly. The eigenmodes of the combined system are <u>polaritons</u>

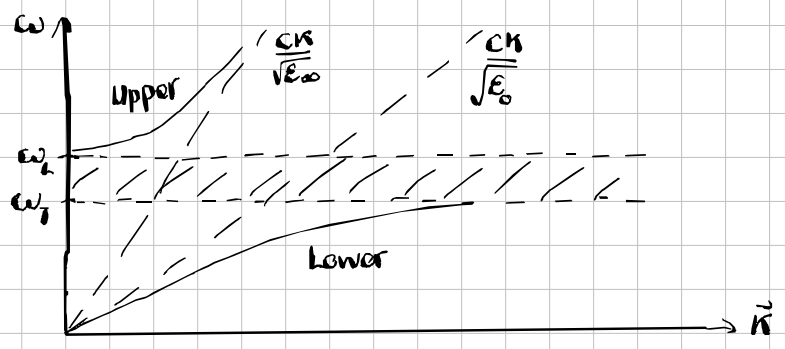
In Lorentz medium,

$$\varepsilon(\omega) = \varepsilon_\infty \frac{\omega_L^2 - \omega^2}{\omega_T^2 - \omega^2}$$ We see $\varepsilon(\omega_L) = 0$ $\varepsilon(\omega_T) \Rightarrow \infty$

Maxwell's equations give the dispersion

$$k^2 = \frac{\omega^2}{c^2}\varepsilon(\omega) \Rightarrow \omega^2(\omega_L^2 - \omega^2) = \frac{c^2k^2}{\varepsilon_\infty}(\omega_T^2 - \omega^2)$$

There are two solutions $\omega_-(k)$ and $\omega_+(k)$



For $\omega_T < \omega < \omega_L$, we have

$\omega_L^2 - \omega^2 > 0$ and $\omega_T^2 - \omega^2 < 0$ $\Rightarrow \varepsilon(\omega) < 0$ $k^2 = \frac{\omega^2}{c^2}\varepsilon(\omega)$??

So k becomes imaginary. This means there is no propagating bulk electromagnetic wave in this range. This is called Restsrahlen band / polariton gap.

At small k, the lower polariton behaves like light in the static dielectric medium.

For frequencies $\omega << \omega_T, \omega_L$, the dielectric becomes

$$\varepsilon(0) = \varepsilon_\infty \frac{\omega_L^2}{\omega_T^2}$$

Using Lyddane-Sachs-Teller $\frac{\varepsilon_0}{\varepsilon_\infty} = \frac{\omega_L^2}{\omega_T^2}$ $\Rightarrow \varepsilon(0) = \varepsilon_0$

Therefore,

$$k^2 \approx \frac{\varepsilon_0\omega^2}{c^2} \Rightarrow \omega_-(k) \approx \frac{ck}{\sqrt{\varepsilon_0}}$$

As $k \to \infty$, $\omega_- \to \omega_T$. The polariton becomes matter-like.

At $k = 0$, $\omega^2(\omega_L^2 - \omega^2) = 0 \Rightarrow \omega_+(0) = \omega_L$

When $\omega >> \omega_L, \omega_T$, the material oscillator cannot follow the field effectively.

$\varepsilon(\omega) \to \varepsilon(\infty)$

$$k^2 \approx \frac{\varepsilon_\infty\omega^2}{c^2} \Rightarrow \omega_+(k) = \frac{ck}{\sqrt{\varepsilon_\infty}}$$

At large k, the upper branch becomes photon like.

Imagine turning the coupling off, we have

$\omega_{photon}(k) = \frac{ck}{\sqrt{\varepsilon_\infty}}$ $\omega_{matter} \approx \omega_T$

The two curves would cross.

$\omega_{photon}(k) = \omega_{matter}$

Instead of crossing, we have hybridized into upper polariton and lower polariton

$$\begin{pmatrix} \omega_{ph} & g \\ g & \omega_{mat} \end{pmatrix} \Rightarrow \omega_\pm = \frac{\omega_{ph}+\omega_{mat}}{2} \pm \sqrt{\left(\frac{\omega_{ph}-\omega_{mat}}{2}\right)^2 + g^2}$$

At resonance, $\omega_{ph} = \omega_{mat} \Rightarrow \omega_\pm = \omega_0 \pm g$ $\Delta\omega = 2g$ (avoided crossing)

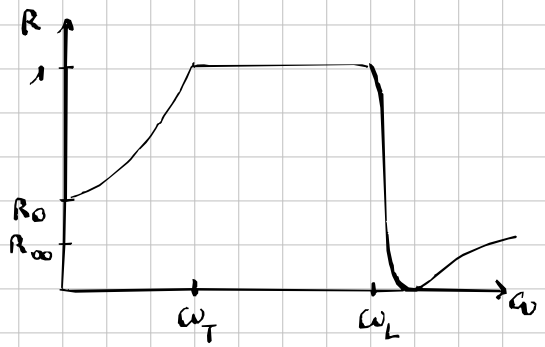
So from

$|LP\rangle \approx |photon\rangle$ and $|UP\rangle \approx |matter\rangle$

This becomes

$|LP\rangle = a|photon\rangle + b|matter\rangle$
$|UP\rangle = -b|photon\rangle + a|matter\rangle$

Reflection



When $\omega_T < \omega < \omega_L$, $\varepsilon(\omega) < 0$. The reflectivity is very high.

At very low frequency $\varepsilon \to \varepsilon_0$

$$R_0 = \left(\frac{\sqrt{\varepsilon_0} - 1}{\sqrt{\varepsilon_0} + 1}\right)^2$$

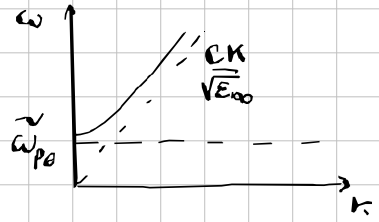
At very high frequency $\varepsilon \to \varepsilon_\infty$

$$R_\infty = \left(\frac{\sqrt{\varepsilon_\infty} - 1}{\sqrt{\varepsilon_\infty} + 1}\right)^2$$

At ω_L, $\varepsilon(\omega_L) = 0$, $R = 1$ so reflectivity does not mathematically jump from 1 to 0 at ω_L.

For a drude model $\omega_0 = 0$ $\quad m\ddot{x} = -eE$

$$\varepsilon(\omega) = \varepsilon_\infty\left(1 - \frac{\tilde{\omega}_{p0}^2}{\omega^2}\right) \Rightarrow \kappa^2 = \frac{\varepsilon_\infty}{c^2}\left(\omega^2 - \tilde{\omega}_{p0}^2\right)$$



The EM transverse dispersion begins at finite frequency

$$\omega(\kappa = 0) = \tilde{\omega}_{p0}$$

Below $\omega < \tilde{\omega}_{p0}$, $\varepsilon < 0 \Rightarrow \kappa^2 < 0$

No propagating

Dispersion Relation in anisotropic media

Generally, $\vec{D} = \vec{\varepsilon}\,\vec{E}$ where

$$\begin{pmatrix} D_x \\ D_y \\ D_z \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

From Maxwell's equations, for a nonmagnetic material with a plane-wave field.

$$\left[\vec{k}\vec{k}^T - k^2 I + \frac{\omega^2}{c^2}\varepsilon_r\right]\vec{E} = 0$$

A non zero field requires $\det M = 0$ dispersion relation

In isotropic case: $M_{ij} = \delta_{ij}\left(\frac{\omega^2}{c^2}\varepsilon - k^2\right)$

Consider uniaxial crystal, the relative permittivity tensor can be written as $\varepsilon_r = \begin{pmatrix} n_o^2 & 0 & 0 \\ 0 & n_e^2 & 0 \\ 0 & 0 & n_o^2 \end{pmatrix}$

This gives two transverse polarization modes.

E_x: $k_x^{mode} = \frac{\omega}{c} n_o$ $\quad E_y$: $k_y^{mode} = \frac{\omega}{c} n_e$

Both modes propagate in z-direction, but their phase velocities differ

$$v_x = \frac{c}{n_o} \qquad v_y = \frac{c}{n_e}$$

This material can be used to perform phase retardation. Imagine incoming light as

$$\vec{E}_{in} = \frac{E_0}{\sqrt{2}}(\hat{x} + \hat{y})$$

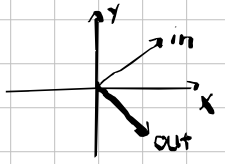
Each component accumulates a phase

$$\phi_x = \frac{2\pi}{\lambda_0} n_o d \qquad \phi_y = \frac{2\pi}{\lambda_0} n_e d \qquad \Gamma = \frac{2\pi}{\lambda_0}(n_o - n_e) d$$

For a half wave plate

$$\Gamma = \pi \pmod{2\pi}$$

$$\vec{E}_{out} \propto \hat{x} - \hat{y}$$



The crystal is not generating new optical frequencies. Instead, it gives two polarization components different phase accumulations. Their recombination produces a different polarization state.